Design of a power supply unit for a CubeSat satellite

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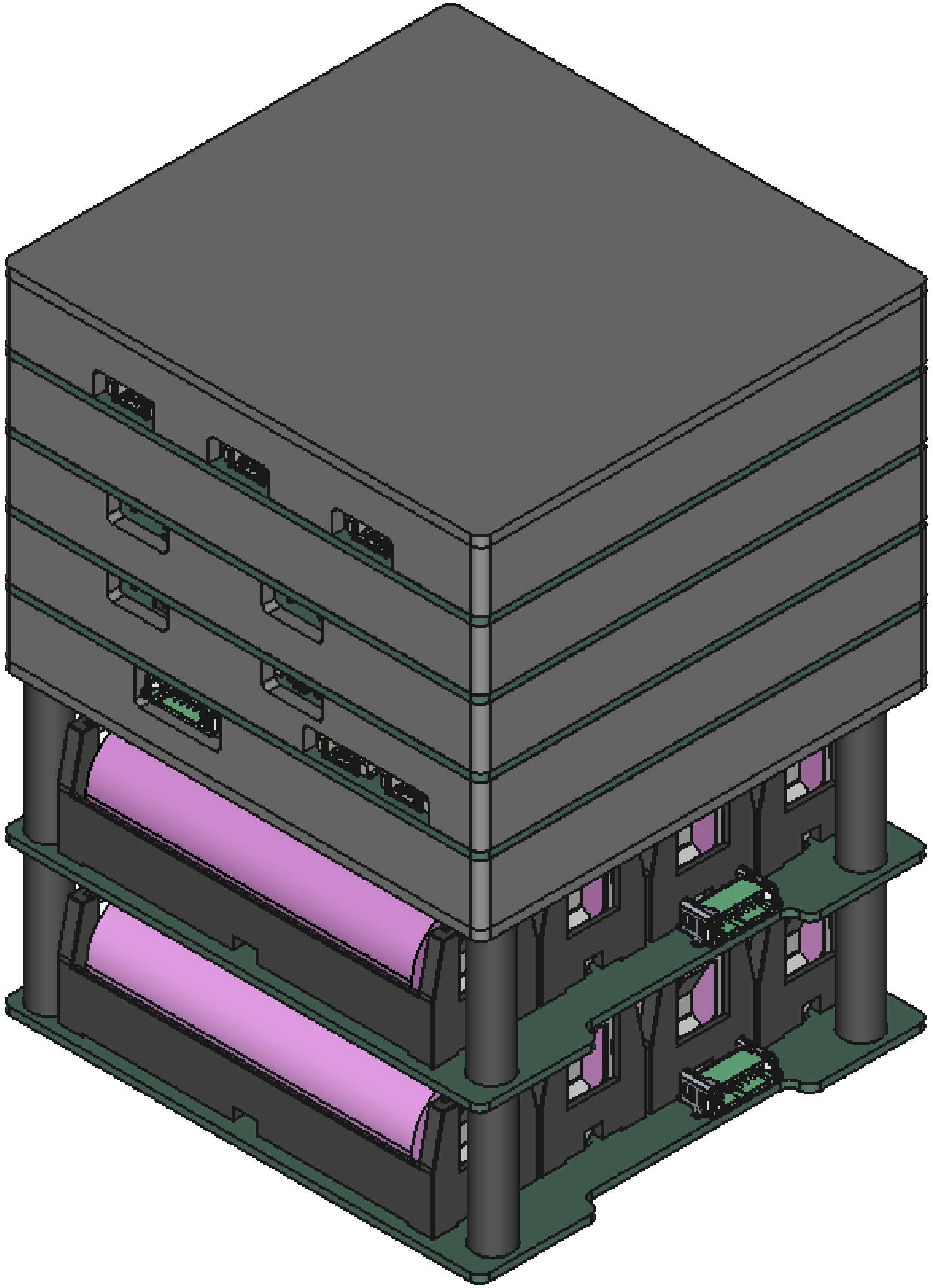
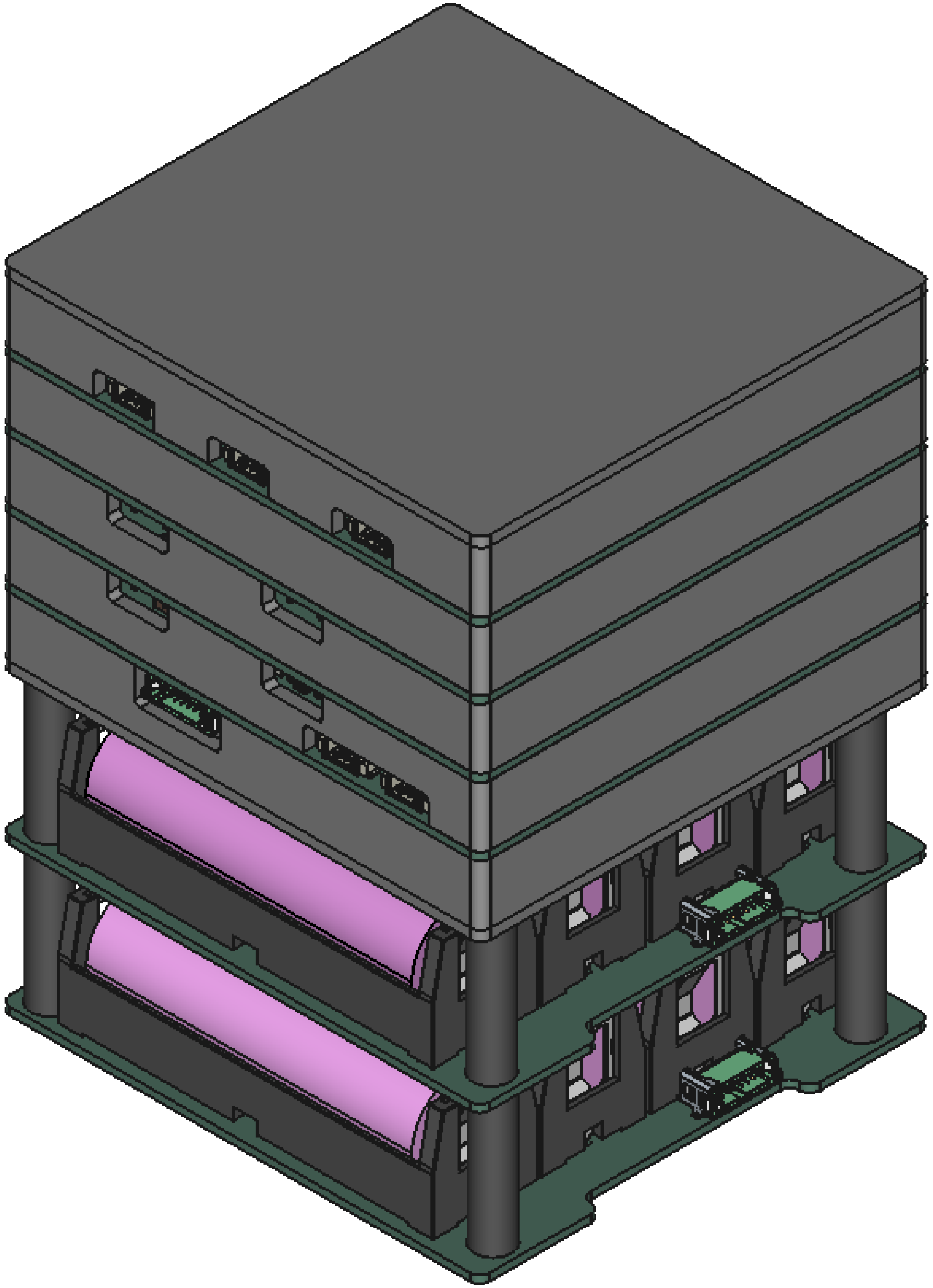


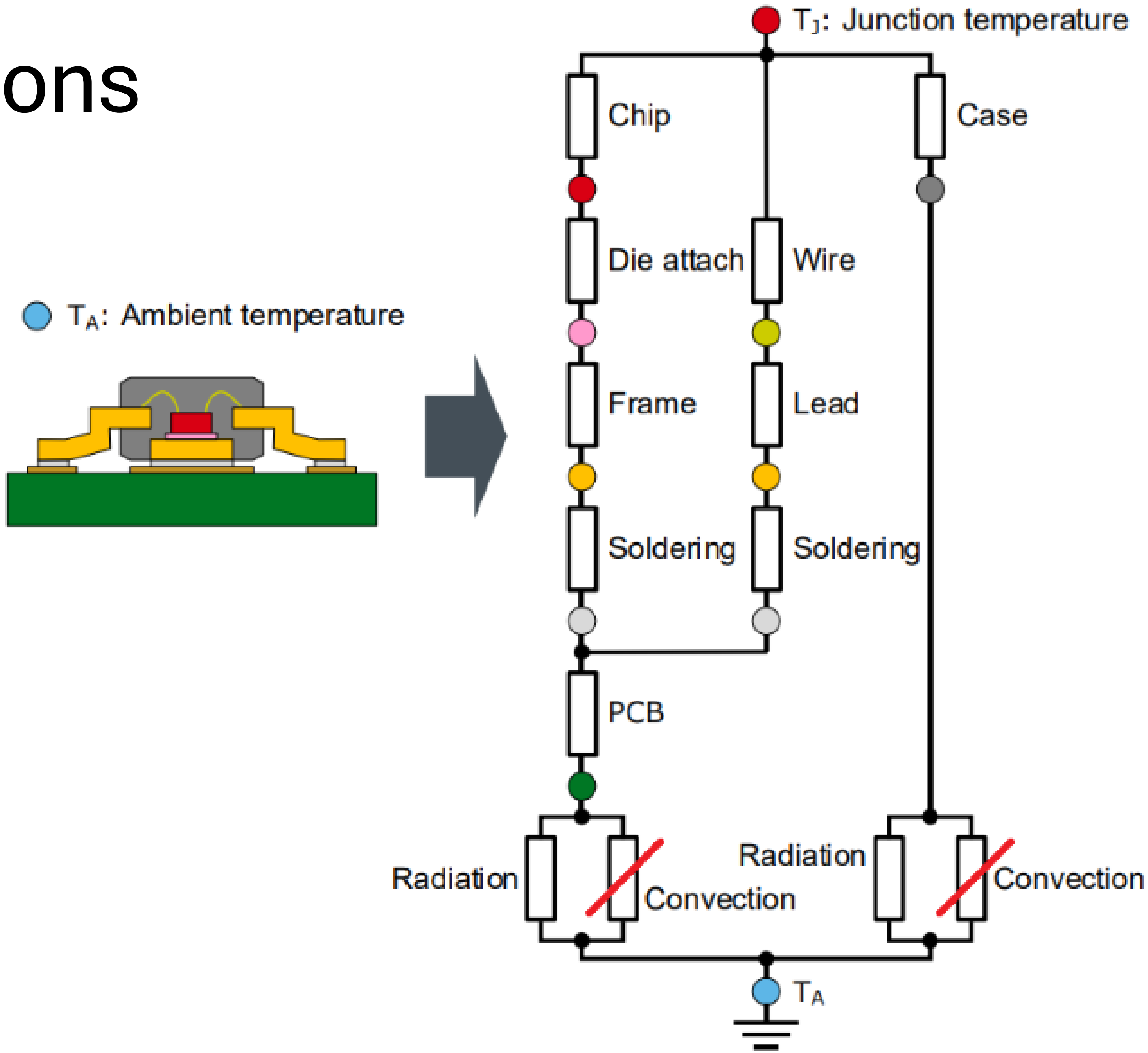
Introduction

- Goal
  - Designing modular eps
- Role of EPS
- Motivation
  - Enhanced reliability
    - For use past LEO
  - COTS components
  - Mission-specific
- 1U – 2U volume ($1U = 100mm^3$)



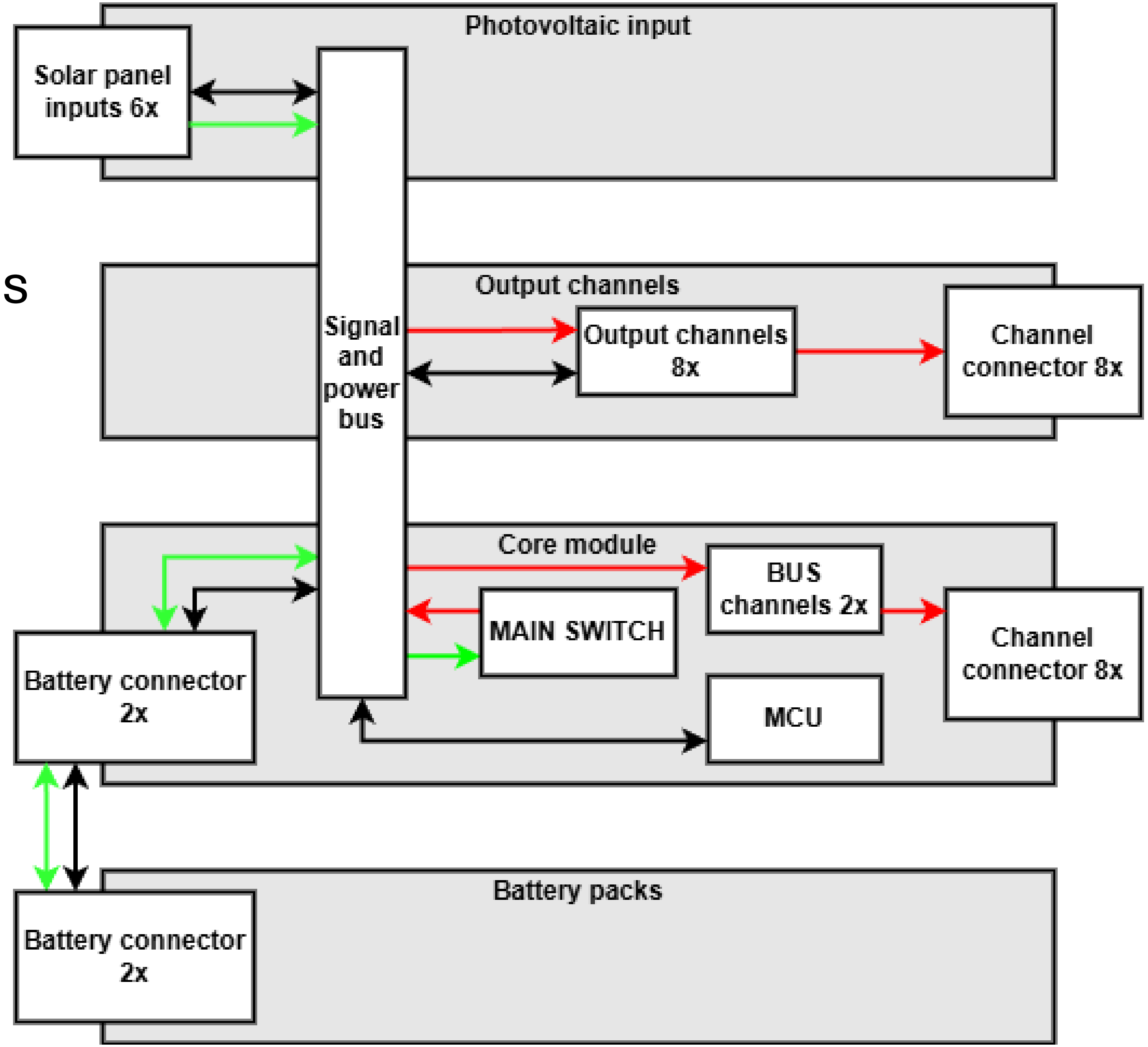
Design considerations

- Space environment
- Heat dissipation
  - Convection
  - Radiation
- Radiation
  - Single event upset
  - Single event transient
  - Latch-up



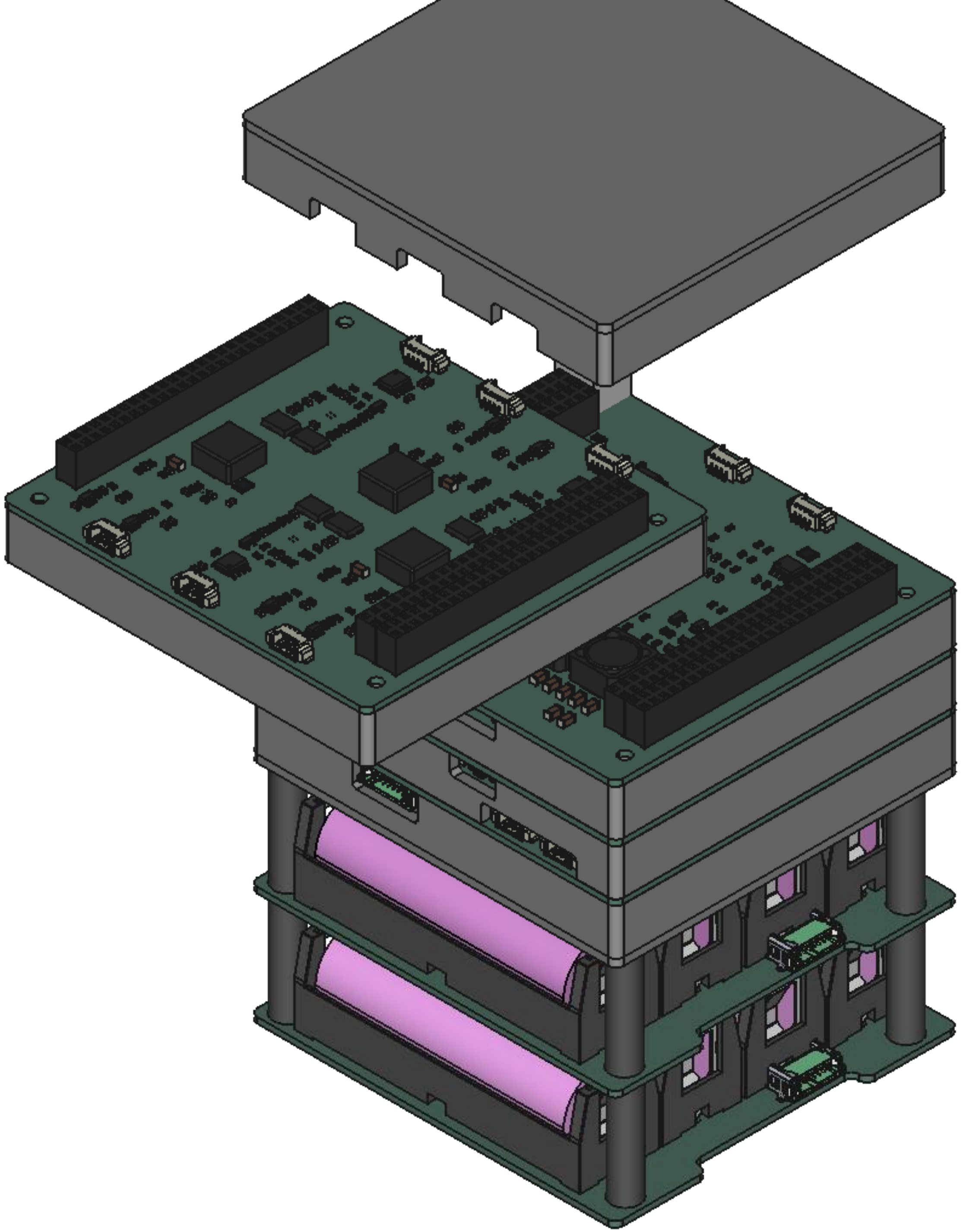
Architecture

- Multiple individual segments
  - Photovoltaic input
  - Output channels
  - Core module
  - Battery packs
- Centralized / Distributed
- Unregulated bus
  - Set to ~16 V
  - 4S batteries (max 16.8 V)



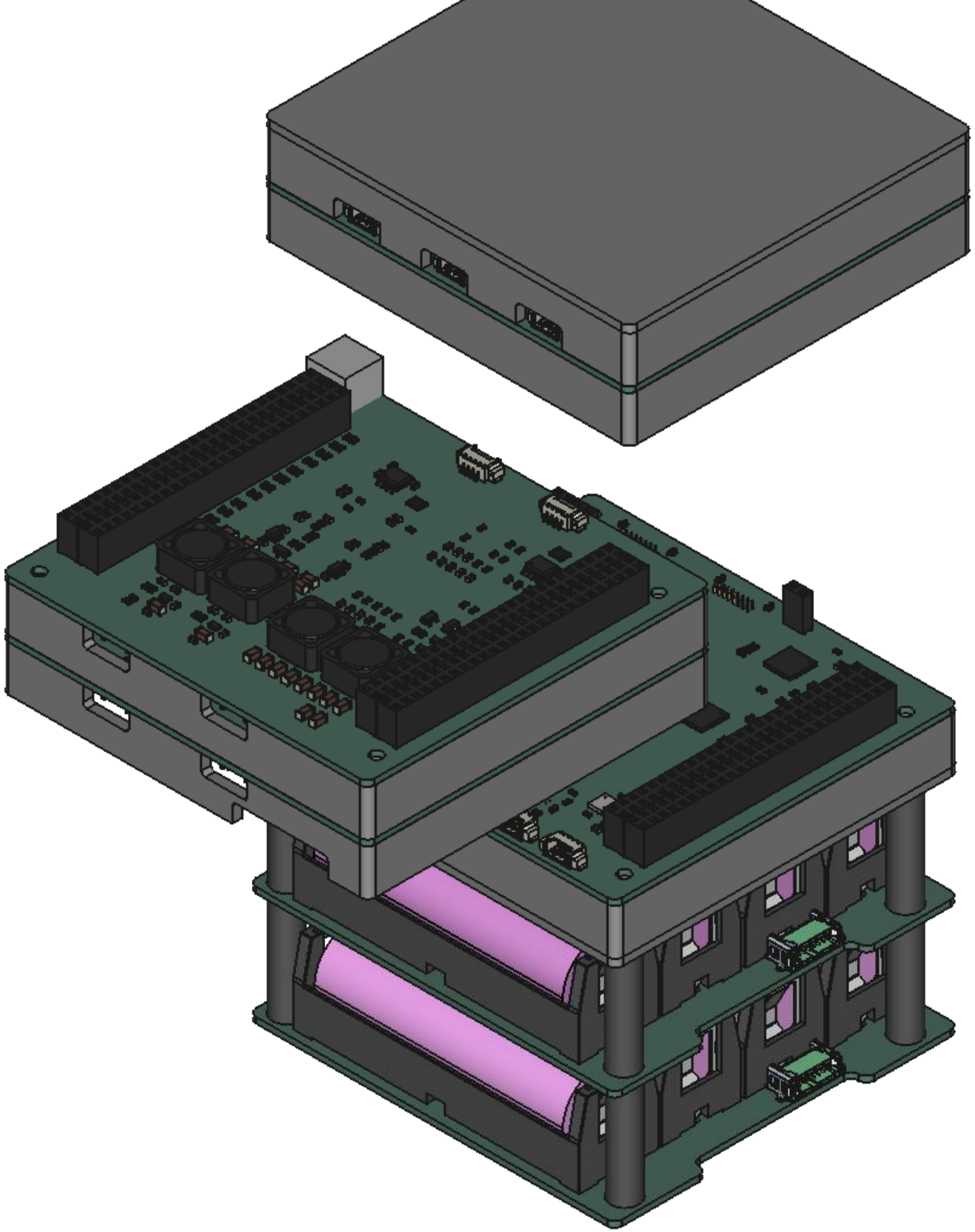
Photovoltaic input

- Includes 6 independent PV inputs
- Each
  - Battery charger
  - Two e-fuse swiches
  - Current and voltage measuremnt
- Fixed point
- MPPT
- 17-27V input voltage



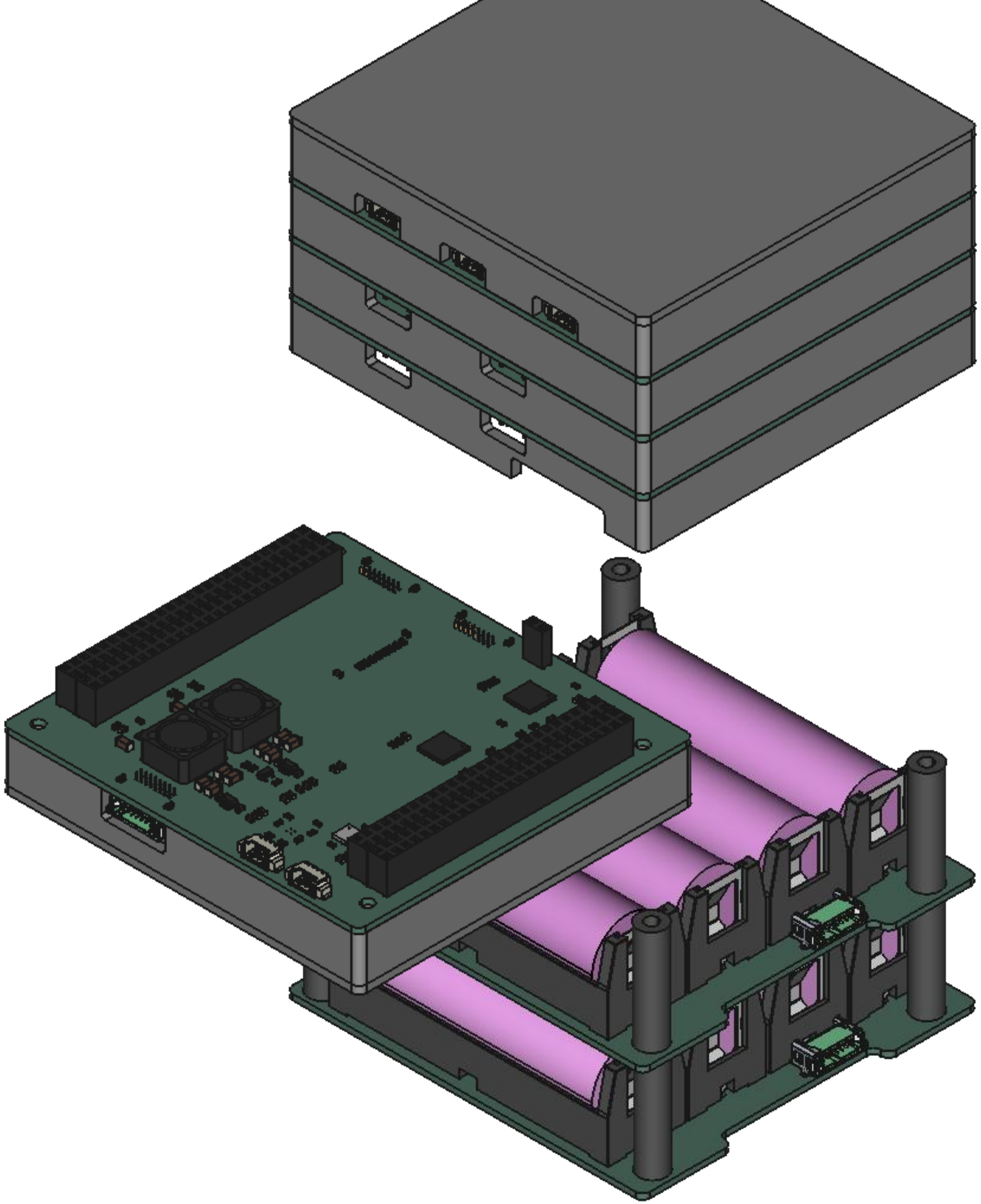
Output channels

- Two identical PCBs
- Each PCB
  - 3 Cold channels (6 total)
  - 1 Hot channel (2 total)
- Each channel
  - Two DC/DC converters
  - Adjustable output voltage
  - Two e-fuse swiches
  - Current and voltage measurement



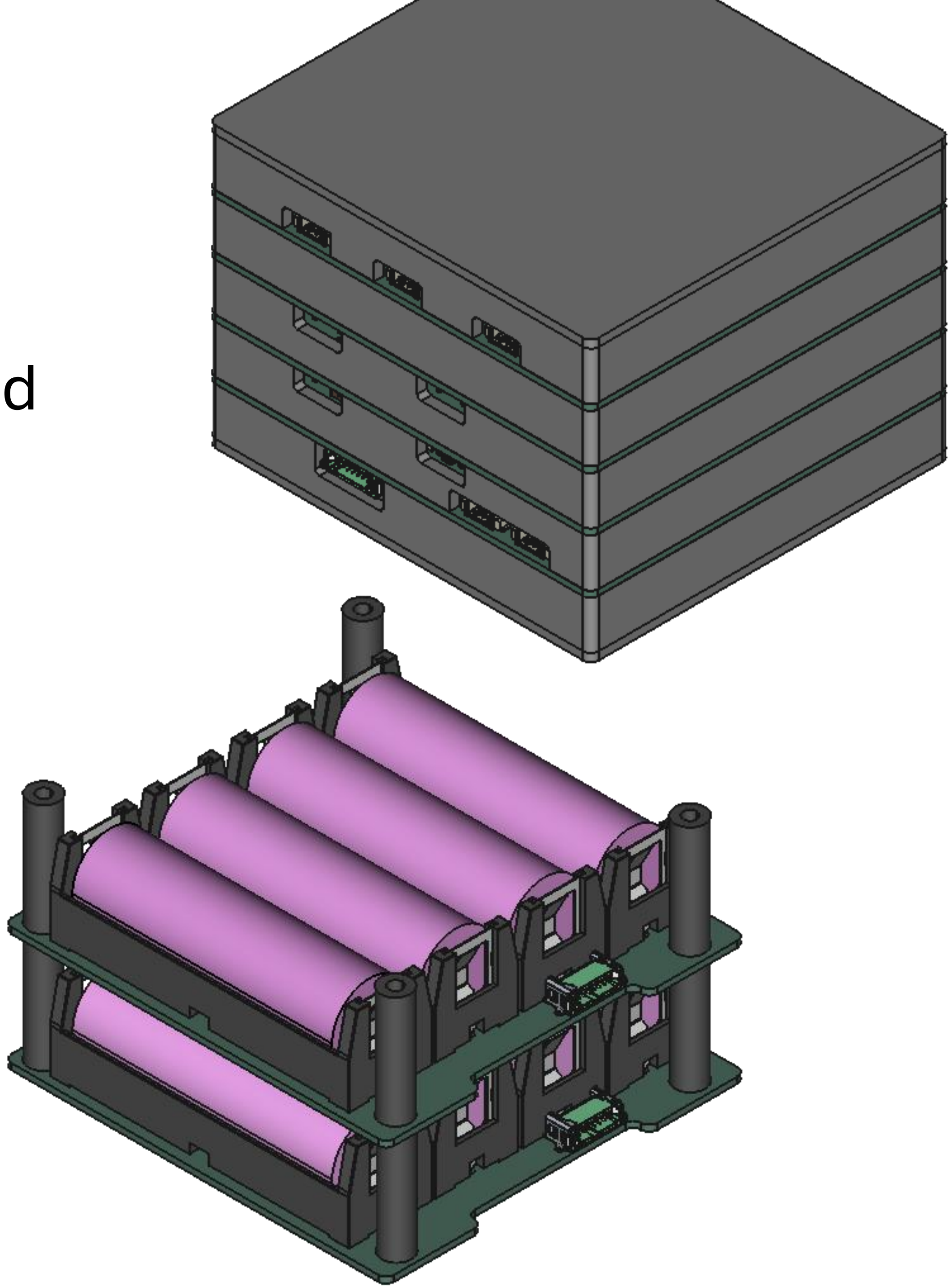
Core module

- Includes the MCU
  - Two MSP430 microcontrolers
  - Two multiplexers
  - CAN and RS422/485 tranceivers
  - SPI, UART, I2C
- Two BUS output channels
  - 2 e-fuse swiches each
  - Current measurement
- Main e-fuse kill / deploy switch



Power storage

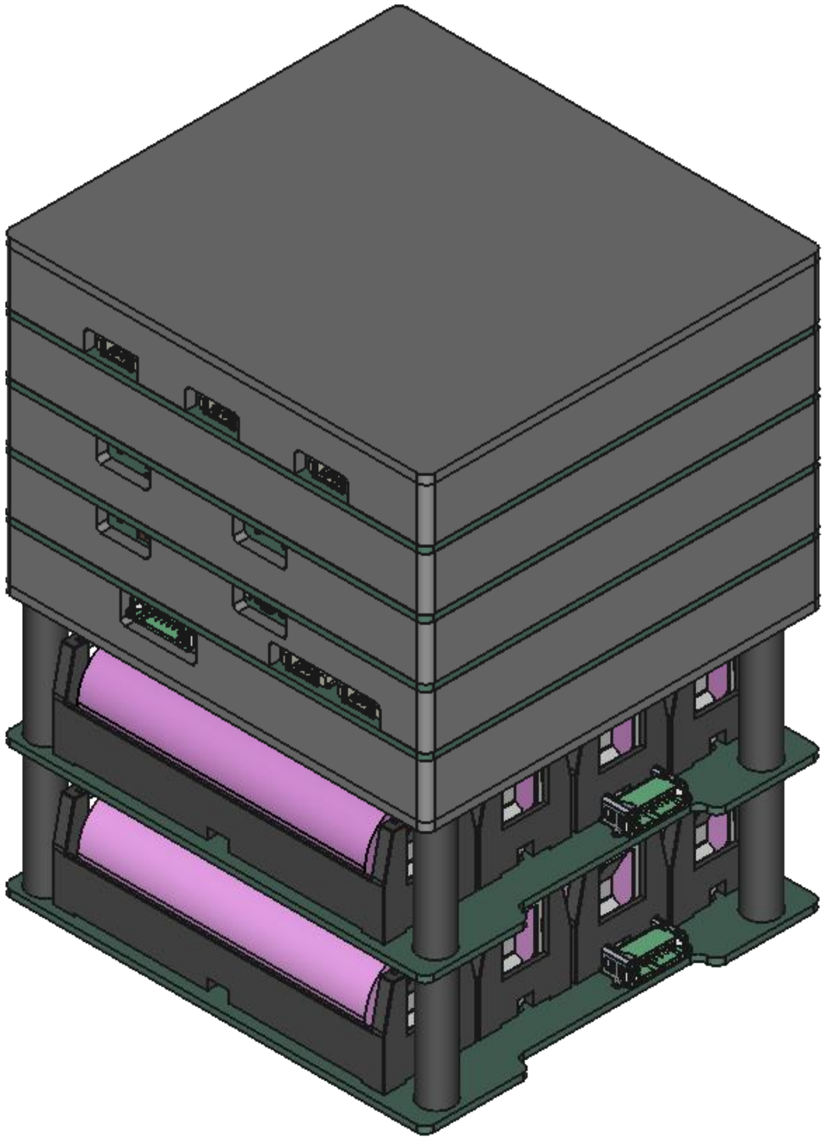
- Two separate battery packs supported
- Each
  - 4S 18650 li-ion batteries
  - BMS Circuit with i2C
  - Temperature measurement
  - Battery heater



Conclusion and future steps

- Test PCB manufactured
  - Simulating EPS architecture
  - Individual parts verified

- Next steps
  - Perform radiation tests on the hardware
  - Pinpoint the weak link
  - Iterate the design



Thank you for your attention!